



Department of Computer Science & Engineering-Data Science

Program Name : B.Tech (CSE-DS)	Academic Session : 2025-26	Semester: 2nd
Course name : Python for Engineers	Course Code: AI102P	Faculty : Prof. Bhagvan Krishna Gupta

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the basics of Python language and control structures to solve computational problems.	Understand	C,P
CO2	Demonstrate the Python complex data types (strings, lists, tuples, dictionaries) and functions for efficient data manipulation and problem-solving.	Apply	C,P
CO3	Apply Object-Oriented Programming concepts and Exception Handling in Python	Apply	C,P
CO4	Apply Python file operations for reading, writing, manipulating files, and processing structured data efficiently.	Apply	P
CO5	Develop simple programs utilizing built-in functions of Python packages like Matplotlib, NumPy, and Pandas.	Create	P,M

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Python for Engineers (AI102P)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	1	1	2	1	1	2	2	-	-	-	3	2
CO-2	3	3	2	1	2	1	1	2	2	-	-	-	3	2
CO-3	3	2	3	1	2	1	1	2	2	-	-	-	3	3
CO-4	3	3	2	1	2	1	1	2	2	-	-	-	3	3
CO-5	3	3	3	1	3	2	1	2	2	-	-	-	3	3
PO Target	3	2.6	2.2	1	2.2	1.2	1	2	2	-	-	-	3	2.6

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

- ❖ The theory courses/ project having credits 3 to 6 should have 5 number of COs. The laboratory course/ mini project/ seminar/ industrial training having credits less than 3 should have 4 number of COs. The Project having 7 to 12 credits should have 6 to 10 number of COs.
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Department of Computer Science & Engineering-Data Science

Program Name : B.Tech(CSE-DS)	Academic Session : 2025-26	Semester: 2nd
Course name : Intro. To Data Science	Course Code: CS101B	Faculty : Dr. Gaurav Dubey

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the foundational concepts of data science and describe its lifecycle and real-world applications.	Understand	C
CO2	Identify different types of data, data sources, and apply basic techniques for data collection and cleaning.	Apply	P
CO3	Use basic Python libraries (NumPy, Pandas, Matplotlib) and spreadsheet tools for simple data analysis tasks.	Analyze	P
CO4	Explain the ethical, legal, and societal implications of data-driven technologies and responsible data handling practices.	Apply	P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Intro. To Data Science (Course Code)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	-	-	-	1	2	-	-	2	-	-	2	-
CO-2	3	2	2	-	-	-	1	-	-	1	-	-	2	-
CO-3	2	3	2	2	2	-	-	-	1	2	1	-	2	-
CO-4	2	1	-	-	-	2	3	3	-	2	-	2	2	-
PO Target	2.5	2	2	2	2	1.5	2	3	1	1.75	1	2	2	-

Signature of Course Coordinator

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Signature of Dean

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Department of Computer Science & Engineering-Data Science

Program Name : B.Tech (CSE-DS)	Academic Session : 2025-26	Semester:2nd
Course name : Data Structure	Course Code: CS201B	Faculty : Dr. Rishab jain

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Use the concept of the array in searching and sorting algorithm.	Apply	C,P
CO2	Illustrate the concept of Dynamic Memory Allocation for Operations on Linked List.	Analyze	C,P
CO3	Analyze different recursion techniques using stack.	Analyze	C,P
CO4	Implementation of Queue and its applications using basic data structures.	Apply	C,P
CO5	Apply the knowledge of tree and binary search tree structures for problem solving.	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Course Name (Course Code)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	1	1	-	1	1	-	-	-	-	-	2	2	-
CO-2	3	2	2	1	1	1	-	-	-	-	-	2	2	-
CO-3	3	2	2	1	1	1	-	-	-	-	-	2	2	-
CO-4	3	2	2	1	1	1	-	-	-	-	-	2	2	-
CO-5	3	2	1	-	1	1	-	-	-	-	-	2	2	-
PO Target	3	1.8	1.6	1	1	1	-	-	-	-	-	2	2	-

Signature of Course Coordinator

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Department of Computer Science & Engineering-Data Science

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Program Name : B.Tech (CSE-DS)	Academic Session : 2025-26	Semester: 2nd
Course name : Computer Organization & Logic Design	Course Code: EC201L	Faculty : Dr. Shipra Srivastava

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Apply the basics of binary arithmetic and codes in digital system design.	Apply	C, P
CO2	Design combinational logic circuits using Boolean functions and gate-level minimization.	Analyze	C, P
CO3	Design sequential logic circuits, including latches, flip-flops, registers, and counters.	Analyze	C, P
CO4	Understand computer organization, including bus architecture, processor organization, and I/O systems.	Understand	C, P
CO5	Understand memory organization, cache, and virtual memory.	Understand	C, P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Computer Organization & Logic Design (EC201L)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	3	2	2	1	1	-	-	-	-	-	-	-	-
CO-2	3	3	2	2	1	1	-	-	-	-	-	-	2	-
CO-3	3	3	2	2	1	1	-	-	-	-	-	-	2	-
CO-4	3	3	2	2	2	2	-	-	-	-	-	-	-	-
CO-5	3	3	2	2	1	1	-	-	-	-	-	-	2	-
PO Target	3	3	2	2	1.2	1.2	-	-	-	-	-	-	2	-

Signature of Course Coordinator

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Department of Computer Science & Engineering-Data Science

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Program Name : B.Tech (CSE-DS)	Academic Session : 2025-26	Semester: 2
Course name : Computer Organization & Logic Design Lab	Course Code: EC201P	Faculty : Dr. Shipra Srivastava

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Apply the basics of binary arithmetic and codes in digital system design	Apply	C, P
CO2	Design combinational logic circuits using Boolean functions and gate-level minimization	Analyze	C, P
CO3	Design sequential logic circuits, including latches, flip-flops, registers, and counters	Analyze	C, P
CO4	Understand computer organization, including bus architecture, processor organization	Apply	C, P
CO5	Understand memory organization, RAM and Read Only Memory (ROM)	Apply	C, P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Computer Organization & Logic Design Lab (EC201P)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	3	2	2	1	1	-	-	-	-	2	2	2	2
CO-2	3	3	2	2	1	1	-	-	-	-	2	2	2	2
CO-3	3	3	2	2	1	1	-	-	-	-	2	2	2	2
CO-4	3	3	2	2	2	2	-	-	-	-	-	2	2	2
CO-5	3	3	2	2	1	1	-	-	-	-	-	2	2	2
PO Target	3	3	2	2	1.2	1.2	-	-	-	-	2	2	2	2

Signature of Course Coordinator

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Department of Computer Science & Engineering-Data Science

Program Name : B.Tech (CSE-DS)	Academic Session : 2025-26	Semester: 2
Course name : Basic Proficiency in Japanese	Course Code: HS103P	Faculty :Prof. Aayush Sharma

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Begin building the foundation in Japanese through sound, rhythm, and simple everyday expressions. Learn how Japanese “feels” when spoken, how to introduce yourself, and how to respond naturally in basic situations.	Understand	C
CO2	Focus on turning words into real communication. Learn how to ask simple questions, share basic information, and understand common replies. Begin forming meaningful sentence patterns that are used constantly in real life.	Understand	C
CO3	Expand beyond greetings and start exploring Japanese used in daily routines. Learn to describe things, talk about surroundings, and express small but important ideas clearly. The unit gently introduces grammar that unlocks a lot of speaking power.	Understand	C
CO4	Focus on strengthening the ability to express smoothly. Learn to speak in longer patterns, understand what's being said more easily, and handle common real-life situations.	Understand	C

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Basic Proficiency in Japanese (HS103P)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	-	-	-	-	-	-	-	-	1	3	-	2	-	-
CO-2	-	-	-	-	-	-	-	-	2	3	-	2	-	-
CO-3	-	-	-	-	-	-	-	-	3	3	-	2	-	-
CO-4	-	-	-	-	-	-	-	-	3	3	-	2	-	-
PO Target	-	-	-	-	-	-	-	-	2.25	3	-	2	-	-

Signature of Course Coordinator

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Department of Computer Science & Engineering-Data Science

Program Name : B.Tech (CSE-DS)	Academic Session : 2025-26	Semester: 2
Course name : Basic Proficiency in German	Course Code: HS104P	Faculty : Ms. Marry Saini

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	To Understand the basic parts of speech of the language, put up basic questions and be able to introduce themselves in respective language and understand the syllables and numbers.	Understand	C
CO2	Understand how language and culture interact in global context and impact intercultural communication.	Understand	C
CO3	Apply their learning in basic conversations, sentence building and understand the social etiquette of professional world	Understand	C
CO4	Utilize the skills of listening, speaking and non-verbal communication in the target language.	Understand	C

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Basic Proficiency in German (HS104P)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	-	-	-	-	-	-	-	-	1	3	-	2	-	-
CO-2	-	-	-	-	-	-	-	-	2	3	-	2	-	-
CO-3	-	-	-	-	-	-	-	-	3	3	-	2	-	-
CO-4	-	-	-	-	-	-	-	-	3	3	-	2	-	-
PO Target	-	-	-	-	-	-	-	-	2.25	3	-	2	-	-

Signature of Course Coordinator

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Department of Computer Science & Engineering-Data Science

Program Name : B.Tech(CSE-DS)	Academic Session : 2025-26	Semester: 2
Course name : Basic Proficiency in French	Course Code: HS105P	Faculty : Ms. Marry Saini

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Introduce themselves in the respective language and understand the syllables and numbers	Understand	C
CO2	Understand how language and culture interact in global context and impact intercultural communication	Understand	C
CO3	Apply their learning in basic conversations and understand the social etiquette of Professional world	Understand	C
CO4	Utilize the skills of listening, speaking and non-verbal communication in the target language	Understand	C

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Basic Proficiency in French (HS105P)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	-	-	-	-	-	-	-	-	2	2	-	2	-	-
CO-2	-	-	-	-	-	-	-	-	2	3	-	2	-	-
CO-3	-	-	-	-	-	-	-	-	1	2	-	2	-	-
CO-4	-	-	-	-	-	-	-	-	3	2	-	3	-	-
PO Target	-	-	-	-	-	-	-	-	2	2.25	-	2.25	-	-

Signature of Course Coordinator

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Department of Computer Science & Engineering-Data Science

Program Name : B.Tech (CSE-DS)	Academic Session : 2025-26	Semester: 2nd
Course name : Basic Proficiency in Spanish	Course Code: HS106P	Faculty : Ms. Shivani Rawat

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Introduce themselves in the respective language and understand the syllables and numbers	Understand	C
CO2	Understand how language and culture interact in global context and impact intercultural communication	Understand	C
CO3	Apply their learning in basic conversations and understand the social etiquette of Professional world	Understand	C
CO4	Utilize the skills of listening, speaking and non-verbal communication in the target language	Understand	C

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Basic Proficiency in Spanish (HS106P)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	-	-	-	-	-	-	-	-	1	3	-	2	-	-
CO-2	-	-	-	-	-	-	-	-	1	3	-	2	-	-
CO-3	-	-	-	-	-	-	-	-	3	3	-	2	-	-
CO-4	-	-	-	-	-	-	-	-	2	3	-	2	-	-
PO Target	-	-	-	-	-	-	-	-	1.75	3	-	2	-	-

Signature of Course Coordinator

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Department of Computer Science & Engineering-Data Science

Program Name : B.Tech (CSE-DS)	Academic Session : 2025-26	Semester: 2nd
Course name : Linear Algebra for Engineers	Course Code: MA103L	Faculty : Dr. Sachin Kumar

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the concept of vector space and subspace.	Understand	C,P
CO2	Apply elementary transformation to solve system of Linear equations.	Apply	C,P
CO3	Understand the concept of complex matrix and Eigen vectors.	Understand	C,P
CO4	Apply the concept of matrix factorization in data decomposition.	Apply	C,P
CO5	Explore the concept of linear transformations and inner product in engineering applications.	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Linear Algebra for Engineers (MA103L)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	3	3	-	-	-	-	1	-	-	-	2	-	-
CO-2	2	2	3	-	-	-	-	1	-	-	-	1	-	-
CO-3	2	2	2	-	-	-	-	1	-	-	-	1	-	-
CO-4	2	2	2	-	-	-	-	1	-	-	-	1	-	-
CO-5	2	2	2	-	-	-	-	1	-	-	-	1	-	-
PO Target	2.2	2.2	2.4	-	-	-	-	1	-	-	-	1.2	-	-

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Department of Computer Science & Engineering-Data Science

Program Name : B.Tech(CSE-DS)	Academic Session : 2025-26	Semester: 2
Course name : Design and Realization	Course Code: ME101B	Faculty : Dr. Pratibha Kumari

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the concept of Computer-Aided Design (CAD)	Understand	C,P
CO2	Apply CAD software to create basic 3D models	Apply	C,P
CO3	Apply CAD and Additive Manufacturing software for 3D printing	Apply	C,P
CO4	Understand the fundamentals of Computer-Aided Manufacturing and CNC machining.	Understand	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Design and Realization (ME101B)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	-	-	-	-	2	-	-	-	-	2	-	3	-	-
CO-2	2	-	2	-	3	-	-	-	-	2	-	3	2	-
CO-3	2	-	2	-	3	-	-	-	2	2	-	3	2	-
CO-4	-	-	-	-	2	-	-	-	2	2	-	3	-	-
PO Target	2	-	2	-	2.5	-	-	-	2	2	-	3	2	-

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